Ascend Adaptation Stack

Serving a 744B MoE on one domestic node

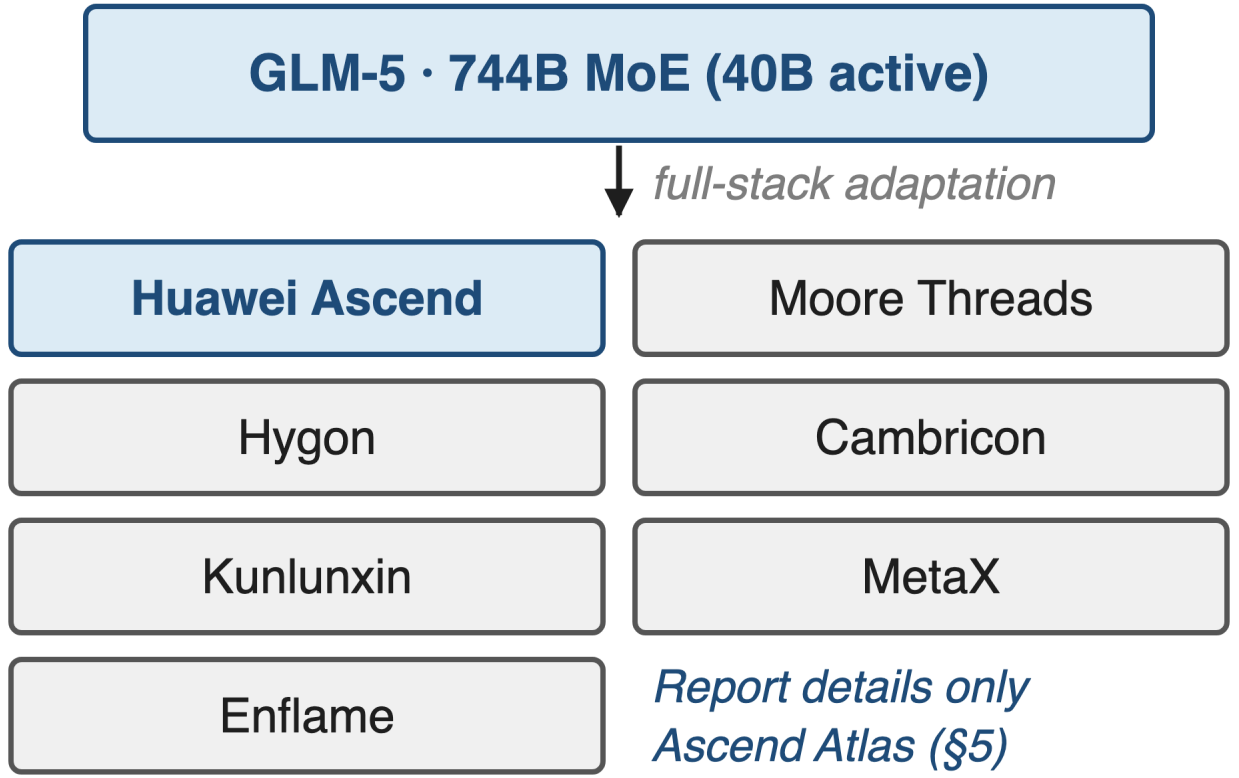
Zhipu AI & Tsinghua · GLM-5

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Frameworks & Platform Stacks · arxiv.org/abs/2602.15763

GLM-5 ships day one on seven domestic platforms

- **744B** MoE, **40B** active, **256** experts
- DSA sparse attention, mid-trained to **200K** context
- Ascend Atlas is the report's worked example



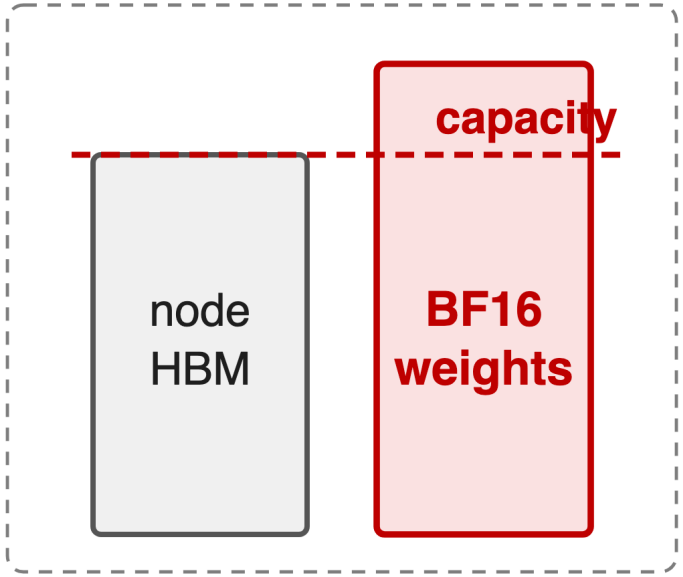
The weights alone outgrow the node

1.4 TiB

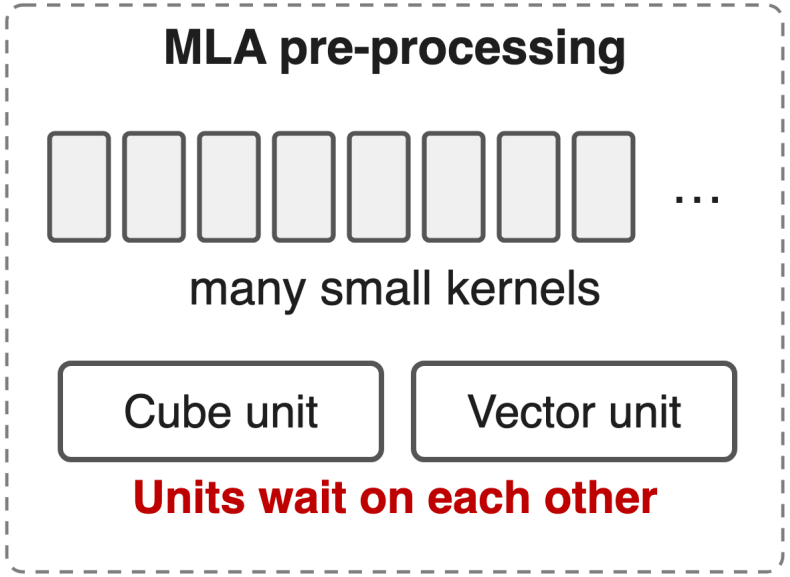
GLM-5 weights at BF16 — one Atlas 800 A3 node holds 1.0 TiB (8 × 128 GB)

Analytic, derived from the architecture in arXiv 2602.15763 Table 10 (p.36); node memory from the vLLM Ascend GLM-5 deployment guide

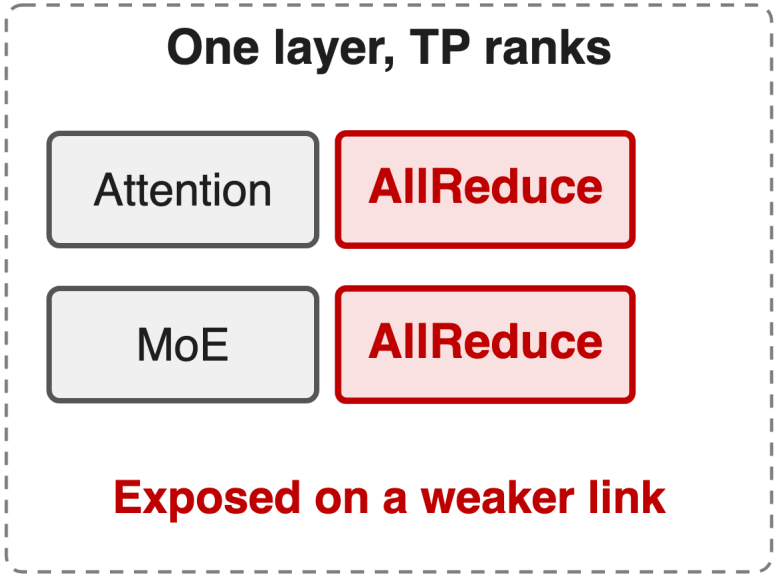
A GPU-shaped stack does not land unchanged



(a) Weights outgrow the node



(b) Kernel swarm, idle units



(c) Collectives on the critical path

Memory, compute units and the fabric run out at the same time

Insight: three layers, three different resources

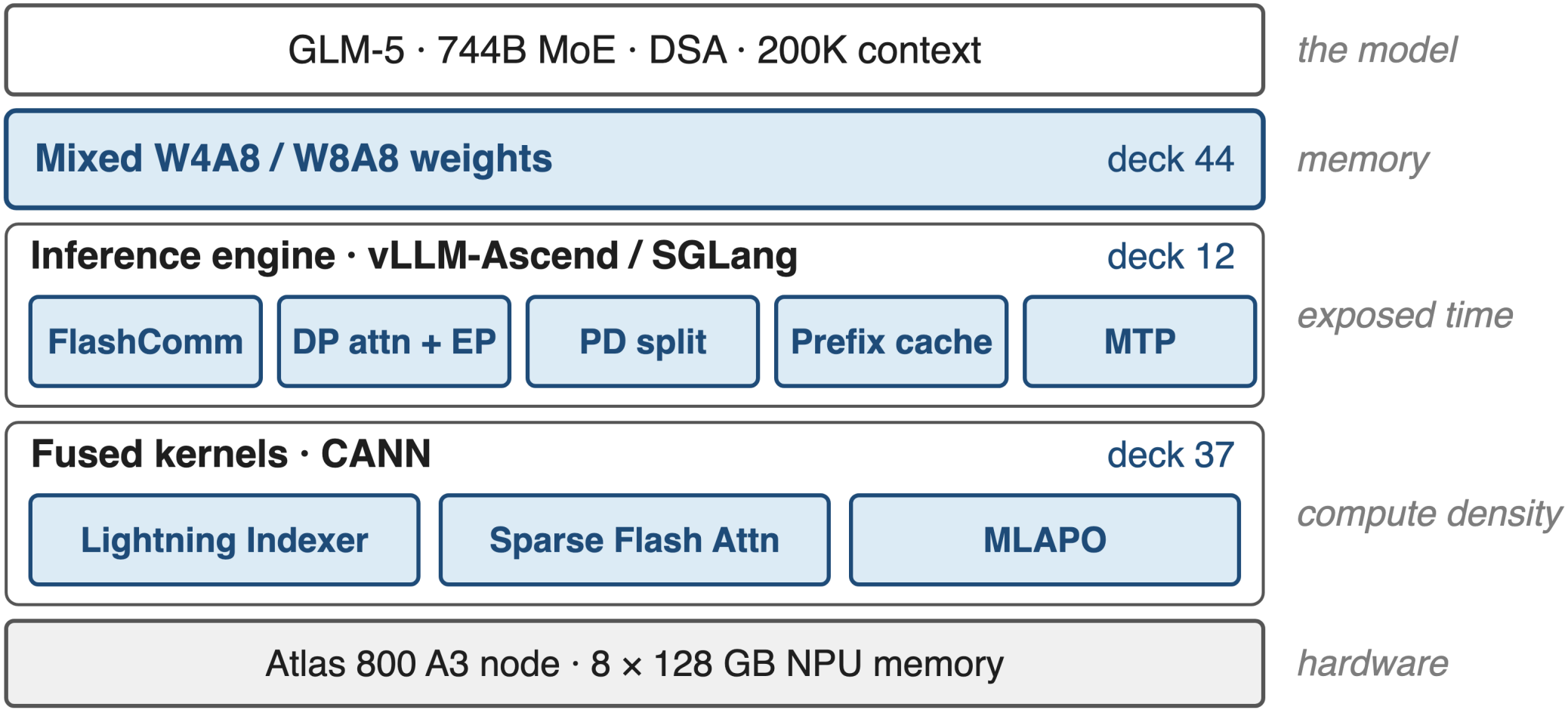
- One port cannot fix memory, compute density and exposed time at once

Insight: three layers, three different resources

- One port cannot fix memory, compute density and exposed time at once

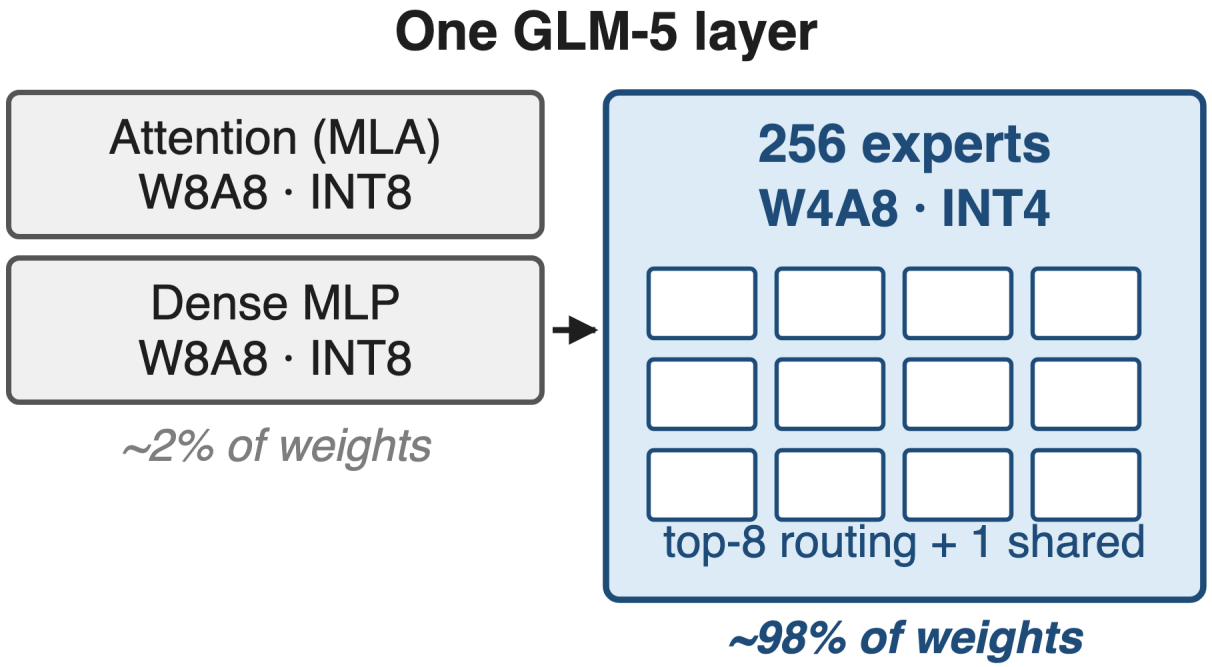
Quantize the weights, fuse the kernels, schedule around what is left

The stack: quantize, fuse, schedule



❶ Mixed INT4/INT8 weights fit one node

- Experts — about **98%** of weights — go to **W4A8**
- Attention and dense MLP stay at **W8A8**
- QuaRot and AWQ-style calibration hold the accuracy

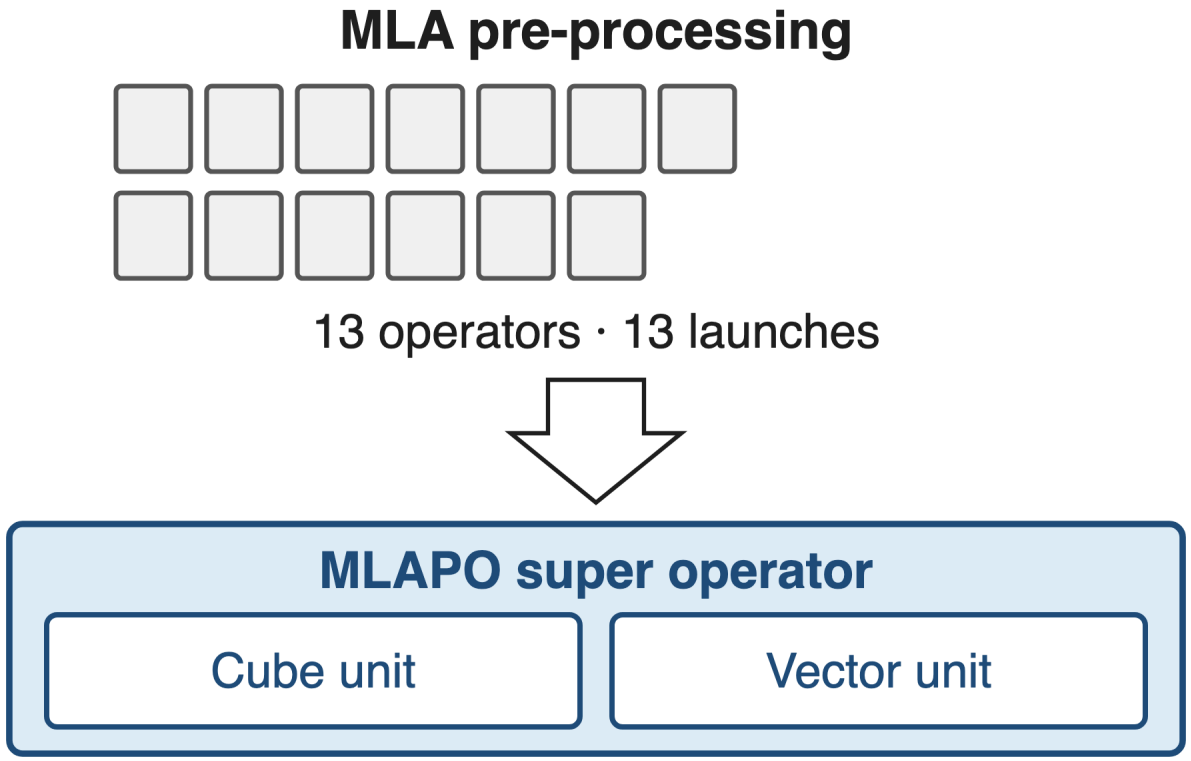


Calibration: QuaRot · Flex_AWQ_SSZ · msModelSlim

Deep dive: deck 44 · expert share derived from Table 10

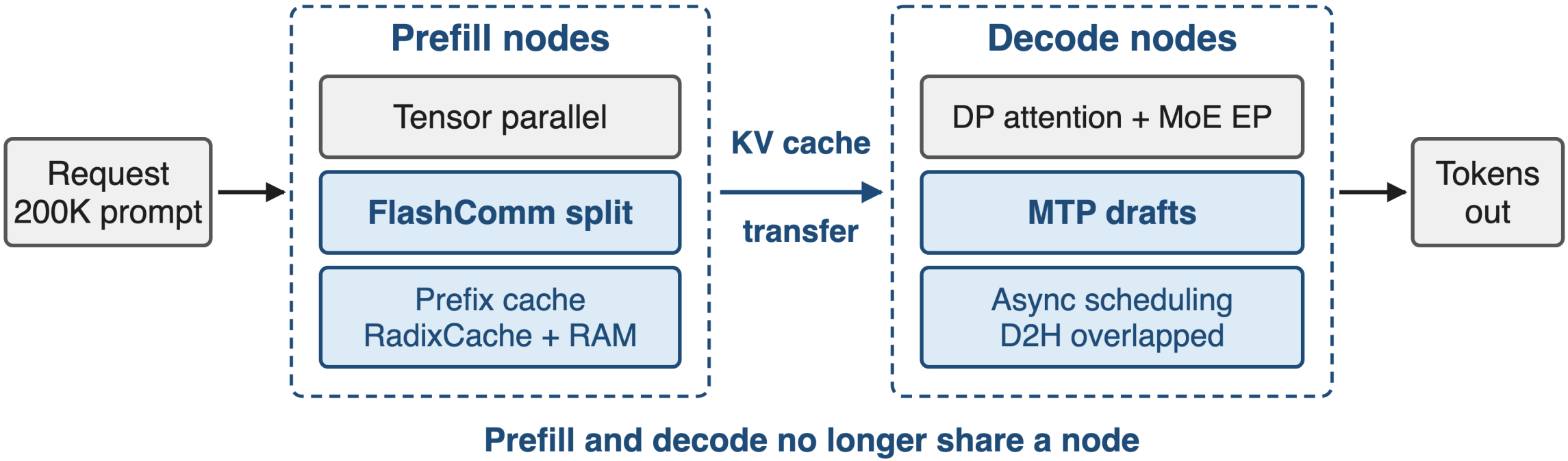
❷ Super-operators replace kernel swarms

- **MLAPO**: **13** pre-processing operators in one kernel
- **Lightning Indexer**: score, ReLU and TopK fused
- **Sparse Flash Attention**: TopK selection beside the attention



Deep dive: deck 37 · the fused kernels of §5

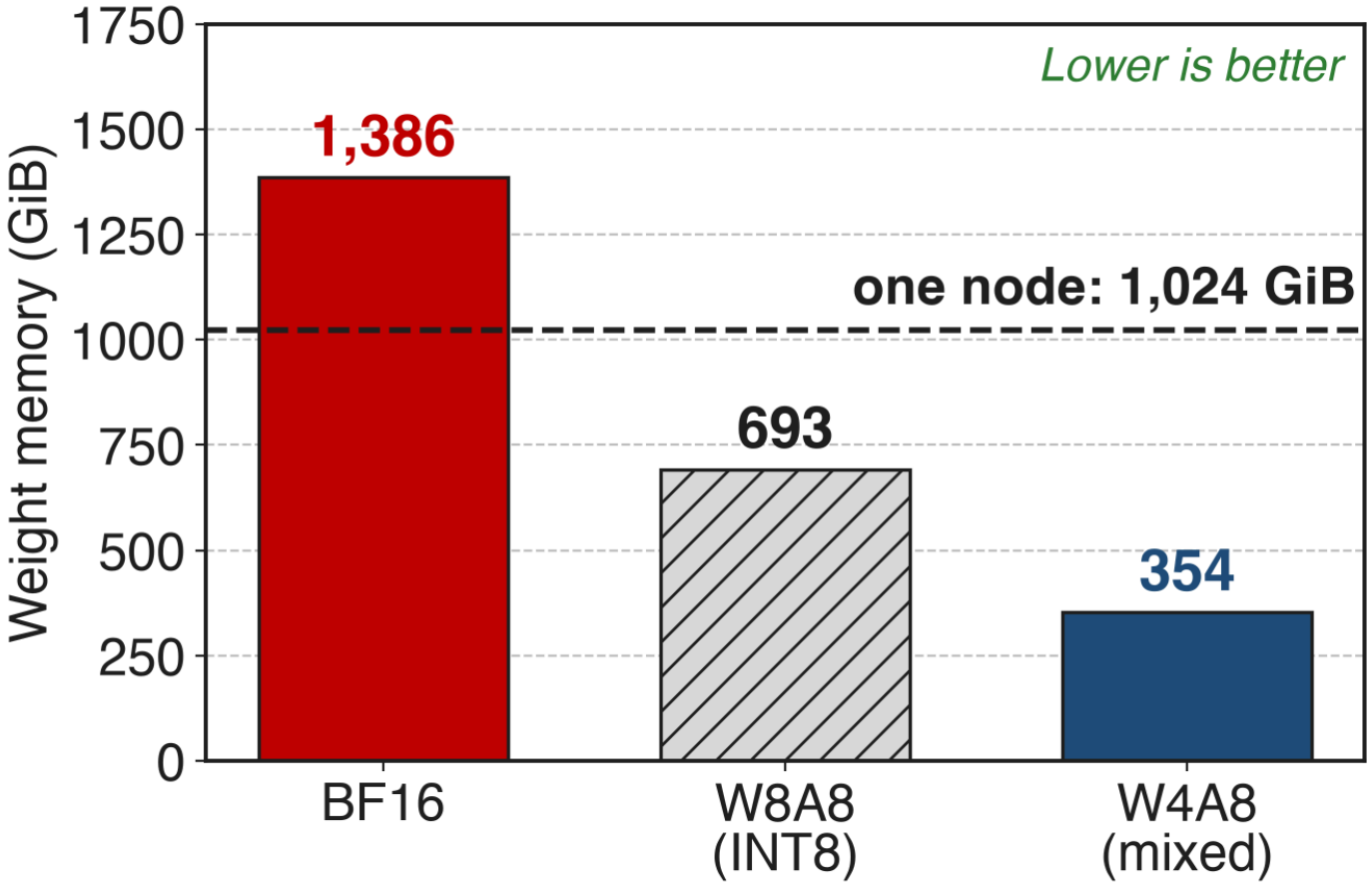
❸ Scheduling hides what is left



Deep dive: deck 12 · FlashComm splits AllReduce; prefix cache, MTP and async scheduling fill the gaps

Only INT4 experts fit 744B on one node

- BF16: **1,386 GiB** against **1,024 GiB** of node memory
- W4A8: **354 GiB**, leaving room for a 200K KV cache



Analytic, from Table 10; excludes quantization scales, KV cache and activations

GLM-5 claims one node and half the cost

- One domestic node reported "comparable to dual-GPU international clusters"
- **50%** lower deployment cost in long-sequence scenarios
- **No baseline GPU, benchmark or metric is named**

Both numbers close §5 with no table behind them

What the report does not tell you

- **Ascend only** of the seven platforms
- **No per-pillar** attribution of the gain
- Deployment only; training hardware unstated
- No accuracy table for the W4A8 weights

Read §5 as an engineering recipe, not as an evaluation

The recipes are upstream already

- vLLM Ascend documents GLM-5 **w4a8** and **w8a8** recipes for one Atlas 800 A3 node
- Reference configs: **198K** context, prefill/decode split, **3** MTP drafts
- Press: a later GLM-5.3-Flash is claimed to run on **100,000** domestic chips

Summary

- GLM-5 weights: **1,386 GiB** at BF16 against **1,024 GiB** per node (analytic)
- **Insight**: memory, compute density and exposed time each need their own layer
- **W4A8** experts · **MLAPO** super-operators · FlashComm and PD scheduling
- Report claims one domestic node and **50%** lower long-sequence cost, unqualified

https://arxiv.org/abs/2602.15763